

Codemix Skill

A plug-in between your AI agent and your action APIs



ALLOW



CONFIRM



BLOCK

*“Order cancel **mat** karo, bas address change karna hai.”*

It understands mixed-language callers, confirms when unsure, and audits every action.

01

Initiative Overview

INITIATIVE

A plug-in between a company's AI agent and its action APIs. It understands mixed-language callers and runs the right API safely.

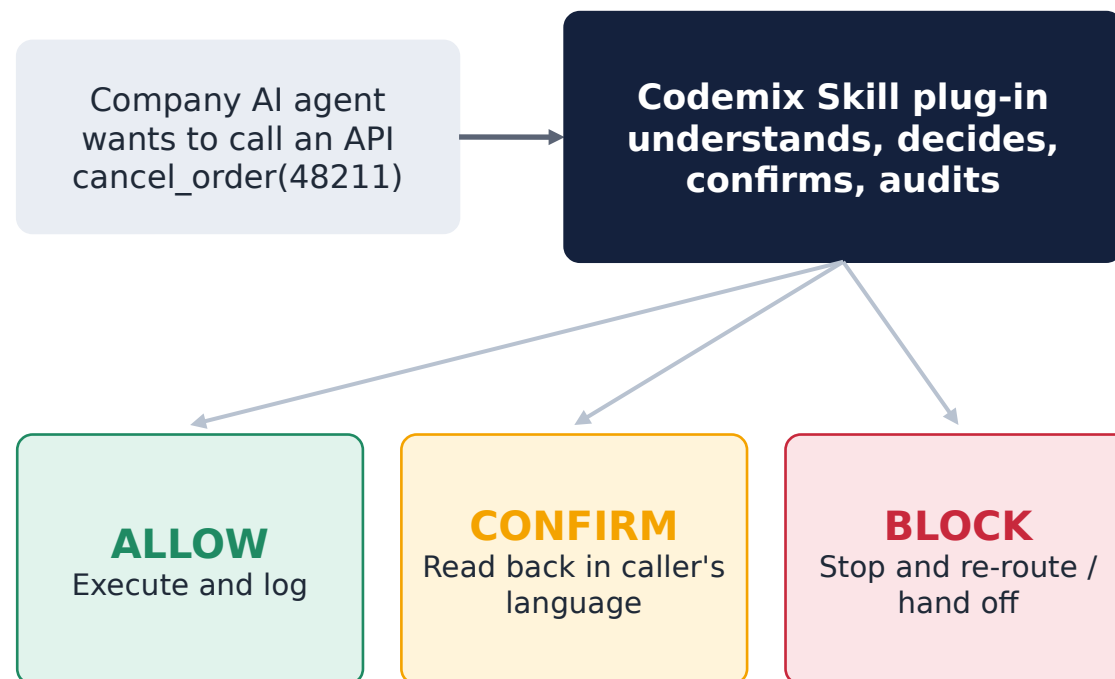
OWNER

Team Ramanathan & Sadhana, builders of Codemix Skill

BUILDS ON

Round-1 code-mix engine + MCP server, live at codemix-skill.vercel.app

How it works



Most actions go straight through. We only ask a question when the action is irreversible and we aren't sure.

Problem Statement

AI support agents now do things, and code-mixed speech is where they will get it wrong

- AI support agents no longer just answer questions. They cancel orders, issue refunds and change addresses on their own.
- Indian callers mix languages mid-sentence, and the meaning often hangs on one small negation word: *mat*, *nahi*, *vendam*, *vaddu*.
- On phone audio, that short word is exactly what speech-to-text mishears or drops.
- The agent then takes the opposite action. It can't be undone, and it costs refunds, reshipping and customer trust.

HINDI NEGATION

Risk: order cancelled

*Order cancel **mat** karo, bas address change karna hai*

Meaning: don't cancel, just change the address

TAMIL NEGATION

Risk: refund issued, no replacement

*Refund **vendam**, replacement anuppunga*

Meaning: I don't want a refund, send a replacement

SPEECH-TO-TEXT DROP

Risk: confident wrong action

Heard: "Order cancel karo..."

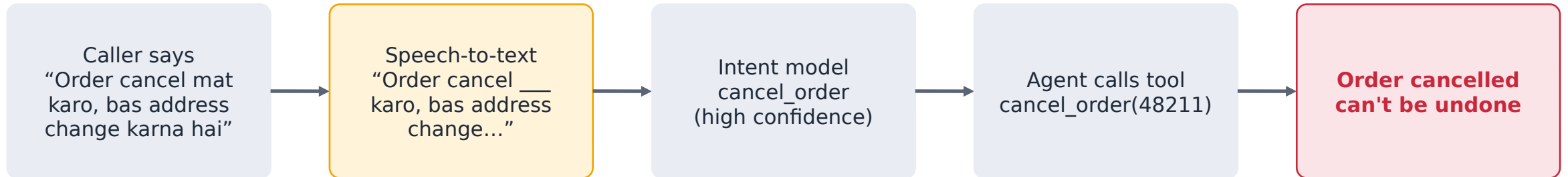
The negation is lost on a noisy line, so the sentence now means the opposite

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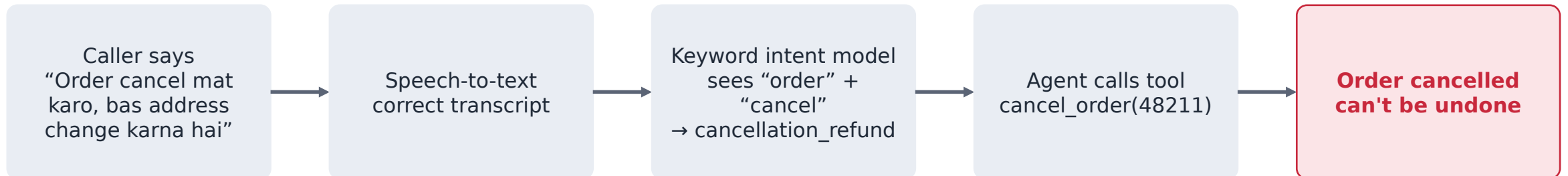
Where It Breaks Today

Two ways the same sentence becomes the wrong action

PATH A: speech-to-text drops the negation



PATH B: transcript is right, but the model ignores "mat"



Measured, not hypothetical: our own Round-1 Codemix Skill engine classifies "Order cancel mat karo, bas address change karna hai" as `cancellation_refund`.

Why Now



Agents now have write access

AI agents, tool calling and MCP let a bot cancel, refund or re-address an order directly. A misunderstanding used to produce a bad answer. Now it produces a bad transaction.



Code-mixed voice is normal in India

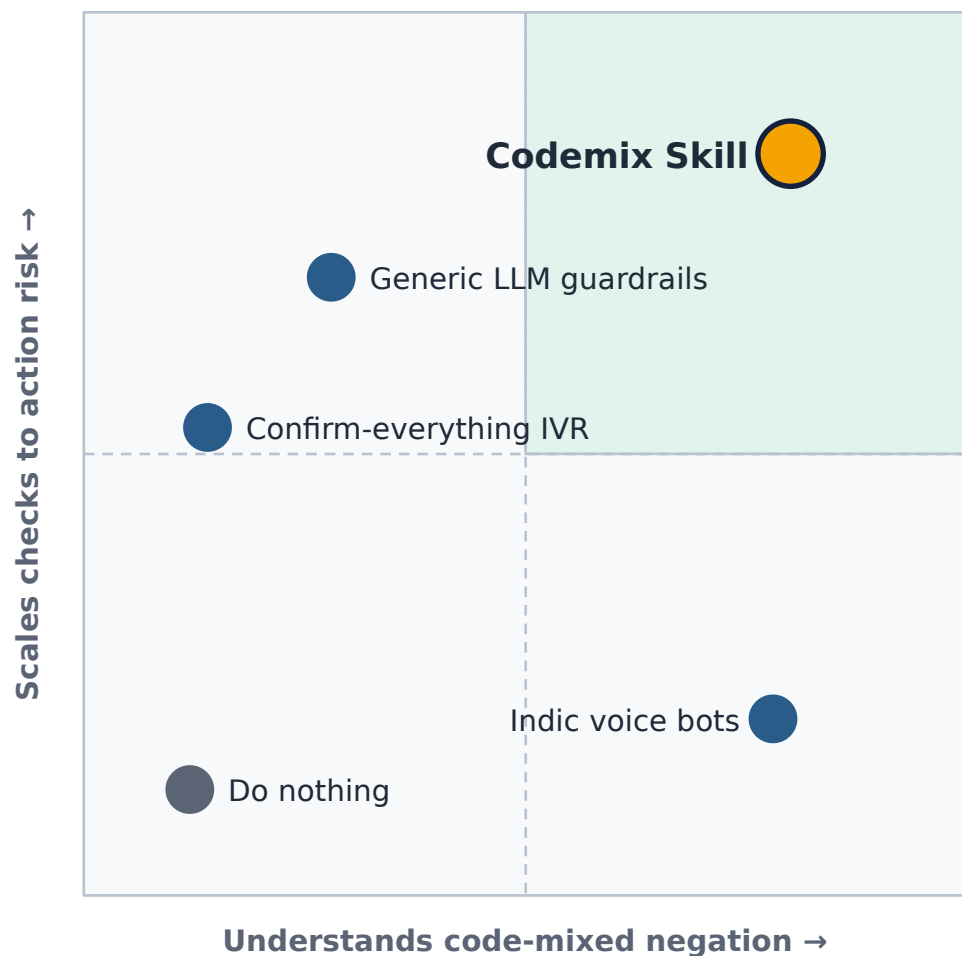
Hinglish and Tanglish callers switch language inside a sentence. Codemix Skill already handles the switching, and negation is the gap we measured in our own engine.



The signals are finally cheap

Speech-to-text now returns word-level confidence (ElevenLabs Scribe), and fast LLMs (Gemini Flash) can check one action inside a live call. Checking every action is now affordable.

Competitive Landscape



Alternative

Strength

Gap

Generic LLM guardrails

Good at policy, toxicity and PII filters

Can't tell that "mat" flips "cancel", and ignores speech-to-text confidence

Confirm-everything IVR

Safe: "press 1 to confirm"

Adds friction to every action, so callers drop off

Indic voice-bot platforms

Strong Indian-language speech recognition

Action safety isn't their layer. Tool calls fire unchecked

Do nothing

Zero effort and zero latency

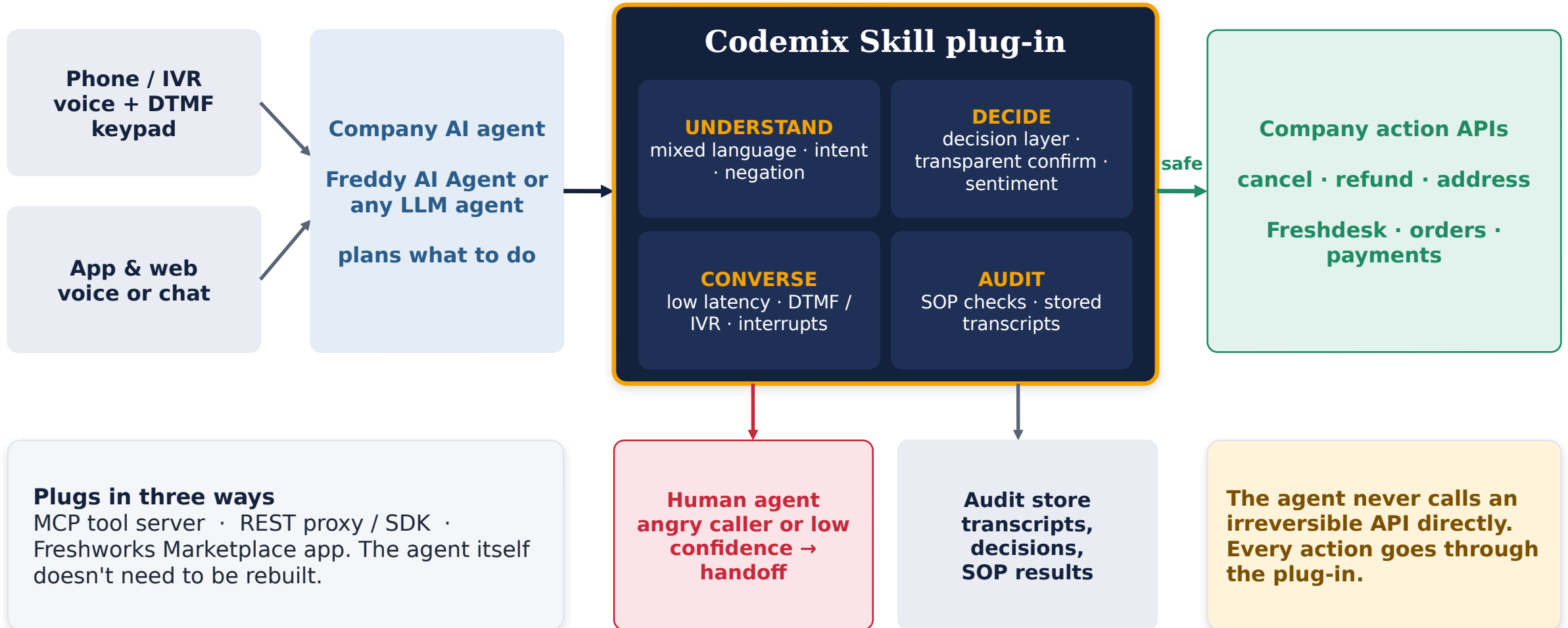
Wrong refunds and cancellations, and customers lose trust

Our edge One plug-in between any AI agent and its APIs: mixed-language understanding, negation, a decision layer and SOP audit, with no rebuild of the agent.

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Proposed Solution: a Plug-in Layer

The company's AI agent calls our plug-in instead of calling its action APIs directly



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What the Plug-in Does

Eleven features in four groups



UNDERSTAND

Mixed languages

Hinglish, Tanglish and more, even when the caller switches mid-sentence

Intent detection

Works out what the caller wants and maps it to the right API tool

Negation handling

Catches mat, nahi, vendam, vaddu that flip the meaning



DECIDE

Decision layer

ALLOW, CONFIRM or BLOCK for every API action

Transparent uncertainty

Tells the caller what it's unsure of and confirms before acting

Sentiment → human

Angry or distressed callers go straight to a person



CONVERSE

Low latency

Fast offline checks; the LLM runs only when a case is unclear

DTMF & IVR

“Press 1 to confirm” works on any phone line and IVR menu

Interrupt handling

The caller can cut in while the agent speaks, and the agent listens



AUDIT

SOP-based auditing

Checks every call against the company's own SOP rules

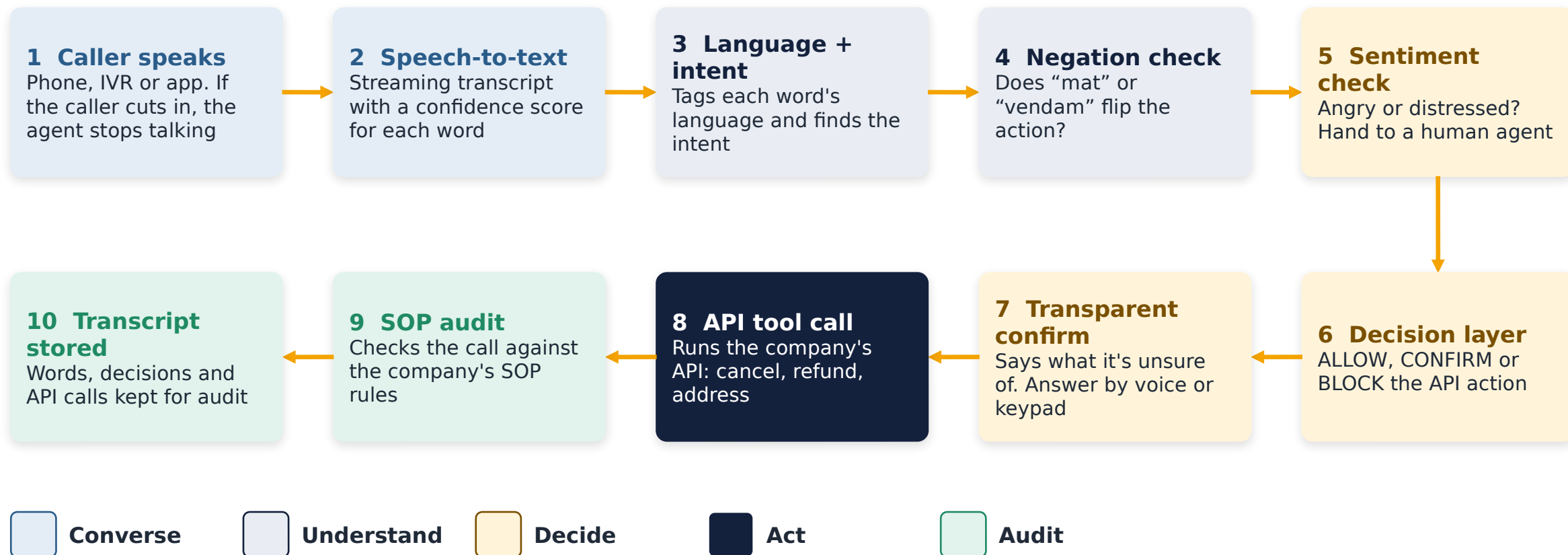
Full audit trail

Stores the transcript, each decision and each API call

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One Call, End to End

Where each feature acts during a single call

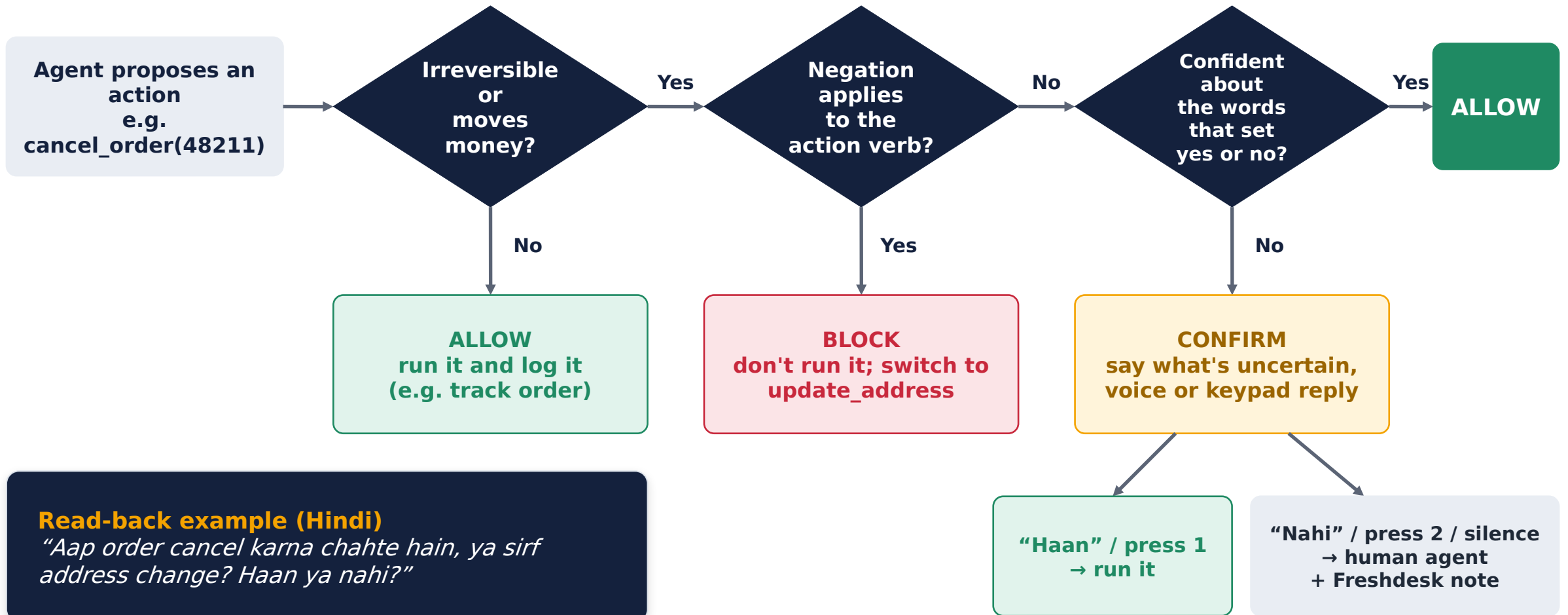


Most calls go 1 → 2 → 3 → 4 → 6 → 8 with no question asked. Steps 5 and 7 only kick in when needed.

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How the Gate Decides

Only the risky, uncertain branch adds friction for the caller



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Confirming Before Acting

The agent says what it's unsure of. The caller answers by voice or keypad, and can interrupt at any time

CALLER

"Refund vendam, replacement anuppunga."

AGENT · spoken in Tamil

"You don't want a refund, you want a replacement for order 48211. I'm not fully sure I heard 'vendam' on this line. Say 'aamaa' or press 1 to confirm. Press 2 to talk to a person."

CALLER · keypad

presses 1

AGENT

"Replacement booked. Reference R-2291."



Transparent uncertainty

Names the exact word or detail it isn't sure of, and how sure it is. No silent guessing on actions that can't be undone.



DTMF & IVR fallback

The keypad works when speech is noisy. It plugs into the company's existing IVR menus: press 1 to confirm, 2 for a person.



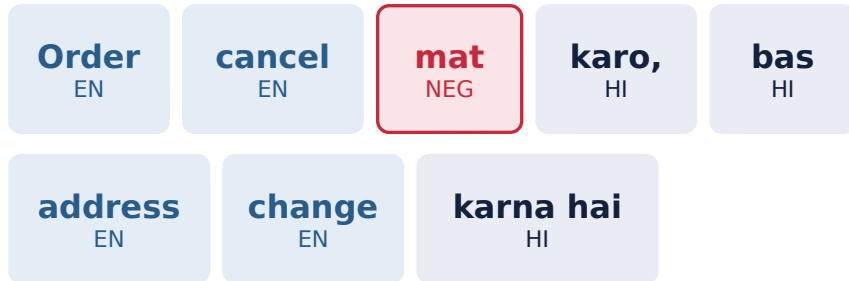
Interrupt handling

If the caller cuts in while the agent is talking, the agent stops speaking and treats the new words as the answer.

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Two Calls Traced Through the Gate

HINDI CALL



Negation “mat” applies to “cancel karo”

Proposed action cancel_order: irreversible

BLOCK cancel_order → run update_address instead (reversible → ALLOW)

TAMIL CALL



Negation “vendam” applies to “refund”. Speech-to-text confidence on it is only 0.58

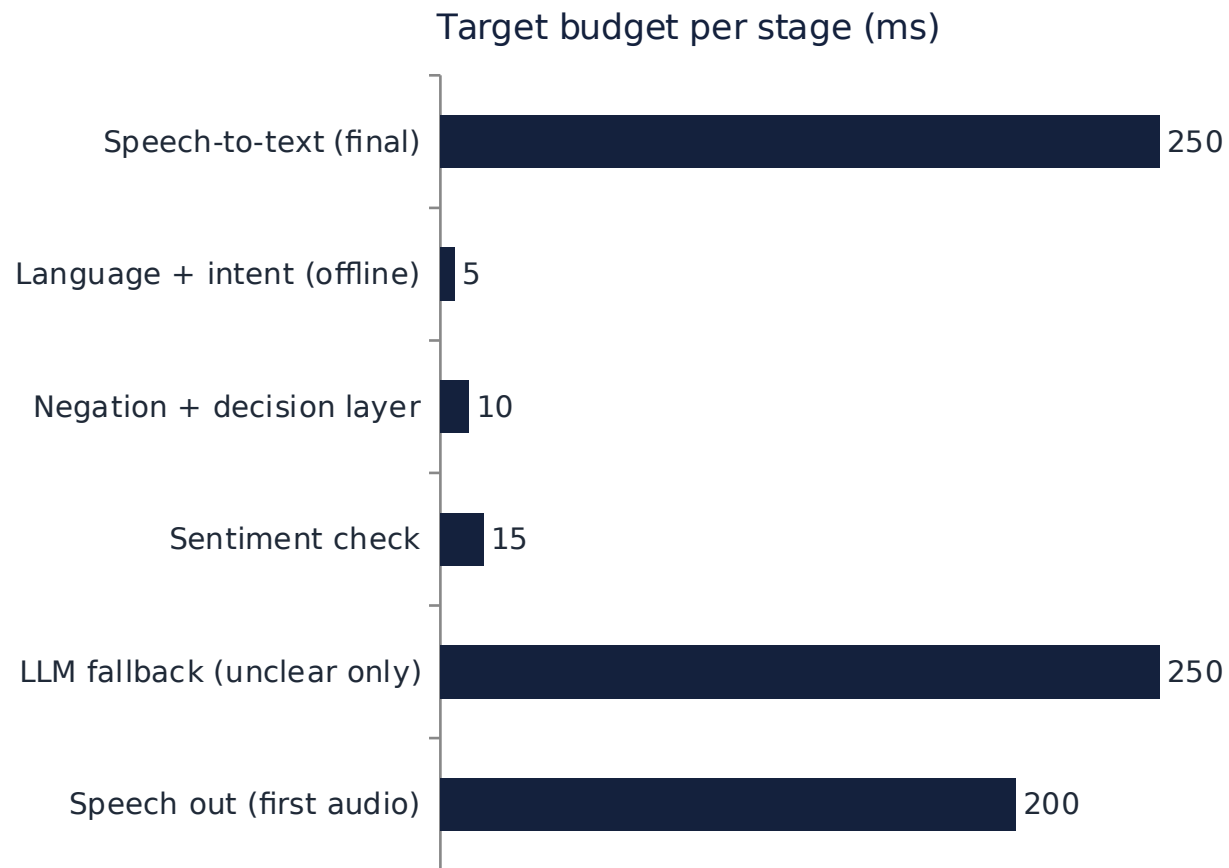
Proposed action issue_refund: moves money

BLOCK issue_refund. **CONFIRM** the replacement: “Replacement anuppalaamaa?”

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Low Latency by Design

Target time budget from the caller's last word to the agent's reply (proposed targets, in ms)



~480 ms

fast path: most calls, no LLM step

< 800 ms

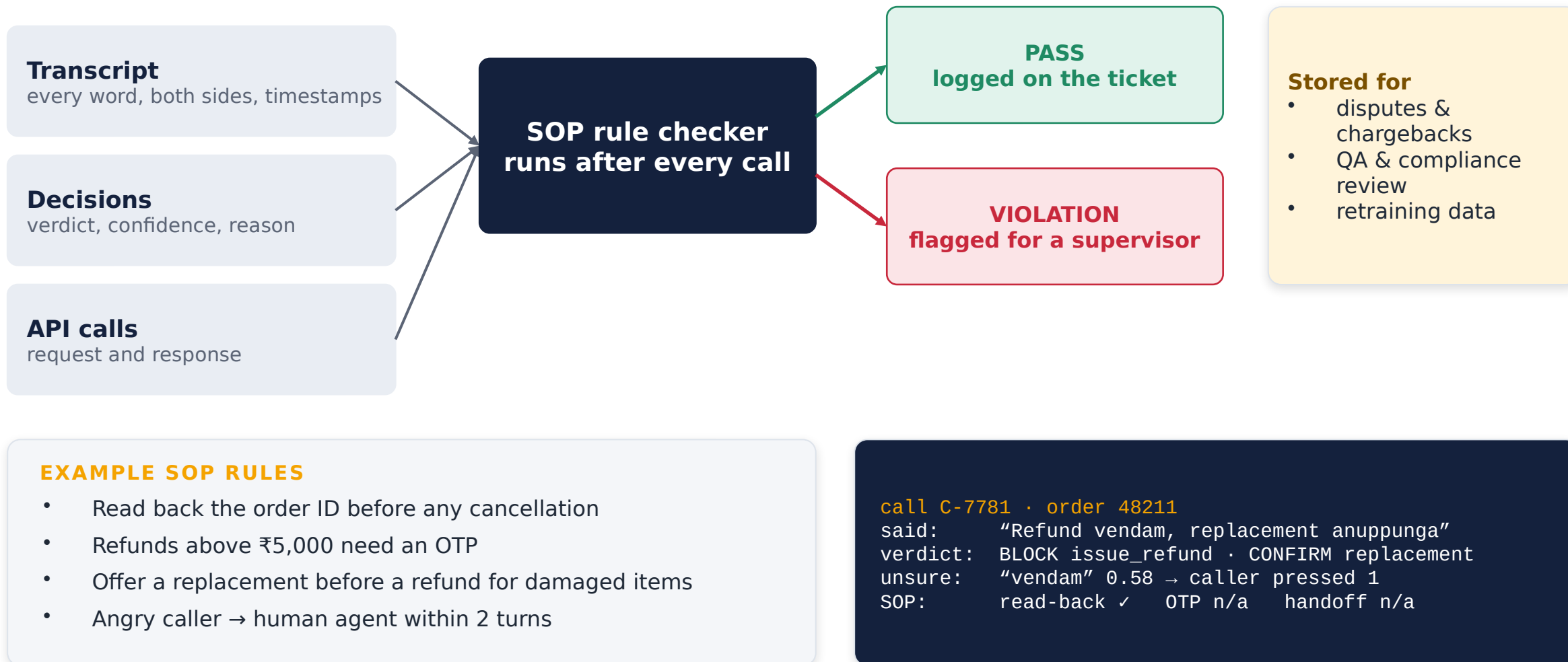
worst case, with the LLM fallback

< 5 ms

offline tagging + intent runs in-process

SOP Auditing & Full Audit Trail

Every call is checked against the company's SOP, and the complete record is stored



Scope: MVP vs Later Phases

PHASE 1 · MVP

Months 0-3

- Plug-in as MCP tool + REST proxy
- Hindi + Tamil + English: intent + negation
- Decision layer + transparent confirm (voice + DTMF)
- Full transcript audit store

PHASE 2

Months 4-6

- Sentiment-based handoff to human agents
- Interrupt handling tuned for phone lines
- SOP rule editor + audit reports
- Adds Telugu, Kannada, Bengali, Marathi

PHASE 3

Months 7-12

- One-click Freshworks Marketplace app
- Chat and WhatsApp channels
- Analytics + learning from human overrides



AI & agentic design

The company's agent plans the tool calls. Codemix Skill is a second, independent agent that checks each one. Fast rule-based checks for each language run first; an LLM is used only for unclear cases, and rules alone still work offline.



Already built (Round 1)

Code-mix tagging engine, intent scoring, MCP server, Freshdesk ticket API, and a live demo on Vercel. Round 2 adds the checks and the gate on top.

Target Customers



PRIMARY

D2C, e-commerce & fintech brands

Indian brands with 20 to 500 support seats, on Freshdesk and Freshcaller, starting to let AI agents cancel, refund and re-address orders for Hindi- and Tamil-speaking callers.



SECONDARY

BPOs and support outsourcers

They run Freshworks for several Indian brands, and one wrong refund costs them a client. They want one policy layer across all their accounts.

Actions guarded in the MVP



Cancel order

Can't be undone once the order leaves the warehouse



Issue refund

Moves money, and reversing it is costly



Change address

Reversible, but a wrong one means a failed delivery

Value Proposition

Plug it in once, and your AI agent can act for mixed-language callers: it gets the intent right, confirms when unsure, hands angry callers to a person and audits every action.

✗ Without Codemix Skill

- Agent acts on its first guess at the intent
- One dropped “mat” leads to a cancelled order
- Angry callers stay stuck talking to a bot
- No transcript or SOP check to prove what happened

✓ With Codemix Skill

- Understands mixed-language callers and runs the right API
- Checks negation and confidence before any irreversible call
- Says what it's unsure of; caller confirms by voice or keypad
- Angry callers reach a person; every call is SOP-audited

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Pricing & Packaging

Proposed. We only charge for the actions we actually guard.

Included

Free

for every Freddy AI Agent customer

- Monitor mode: flags risky actions, never blocks
- Hindi + English
- Weekly wrong-action report

Codemix Skill Add-on

\$5

per 1,000 guarded actions

- Full plug-in: decision layer, DTMF confirm, interrupts
- Sentiment handoff + SOP audit
- All MVP languages
- First 30 days free

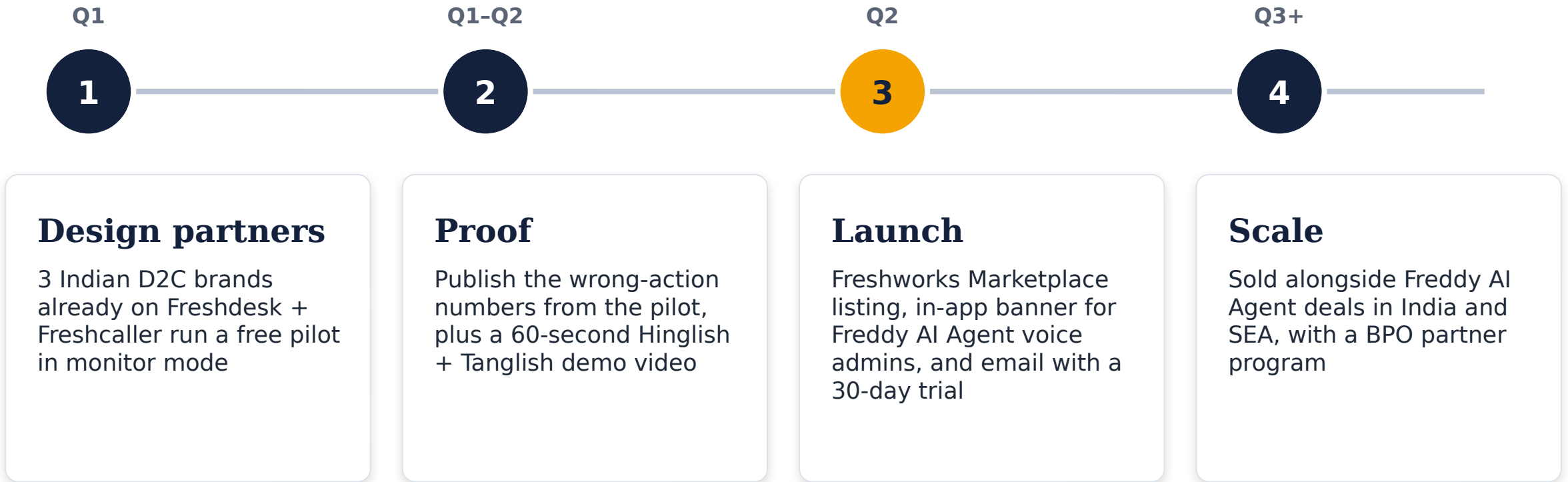
Enterprise

Custom

annual contract

- Custom SOP rule packs
- Transcript retention + data residency in India
- SLA

09 Go-to-Market Plan



Free monitor mode shows each customer their own wrong-action count, and that report is what sells the paid add-on.

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Costs & Resources Needed



3 engineers

plug-in core, voice / IVR, Freshworks app



1 linguist / data annotator

Hindi + Tamil negation set



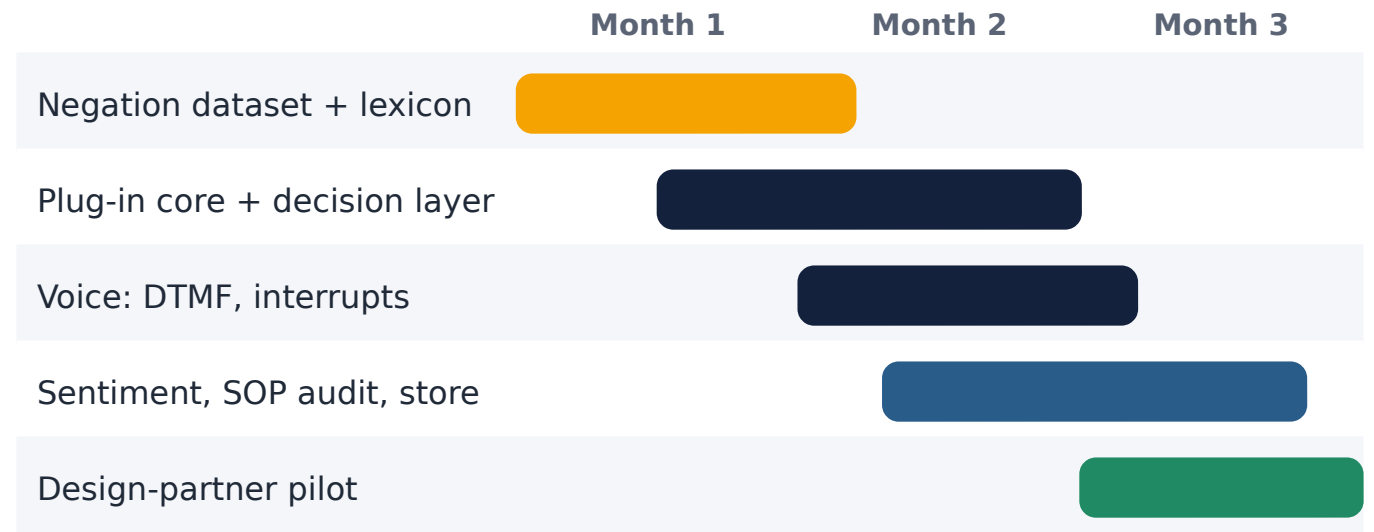
Part-time PM

pilots + pricing

~\$90K

to MVP in 3 months: salaries, speech-to-text and LLM usage, and labelling ~5,000 negation utterances

MVP timeline



Optional: ARR estimate (assumption-based)

100 paying customers × 20,000 guarded actions / month × \$5 per 1,000 = \$10K MRR, or about **\$120K ARR in Year 1**. Every number here is an assumption to test in the pilot.

Expected Benefits & Success Metrics

< 0.5%

Wrong actions

irreversible API calls on negated Hindi + Tamil requests

< 150 ms

Added latency (p95)

what the plug-in adds on top of the agent

≤ 15%

Confirmation rate

share of guarded actions that ask a question

100%

Audit coverage

API actions stored with transcript + SOP result

−30%

Reversal tickets

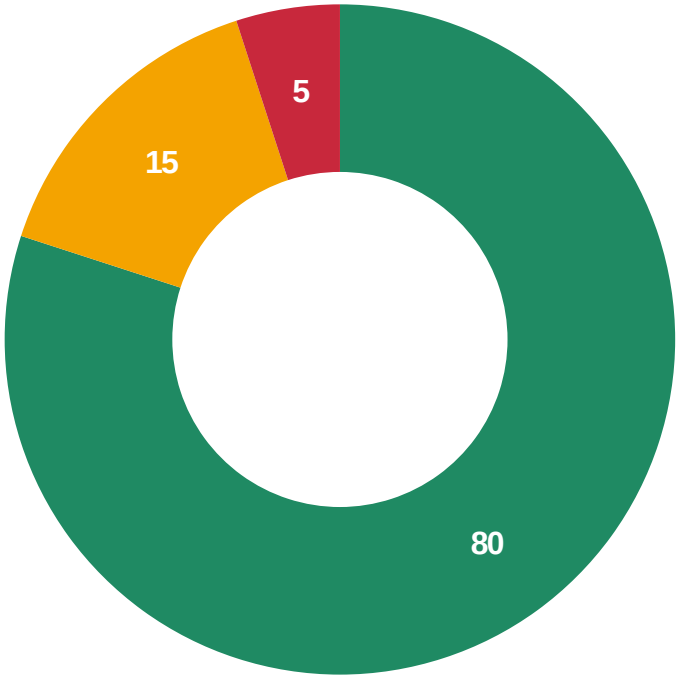
refund / cancellation reversals at pilot customers

15%

Adoption

of eligible voice customers within 2 quarters

Target mix of guarded actions (%)

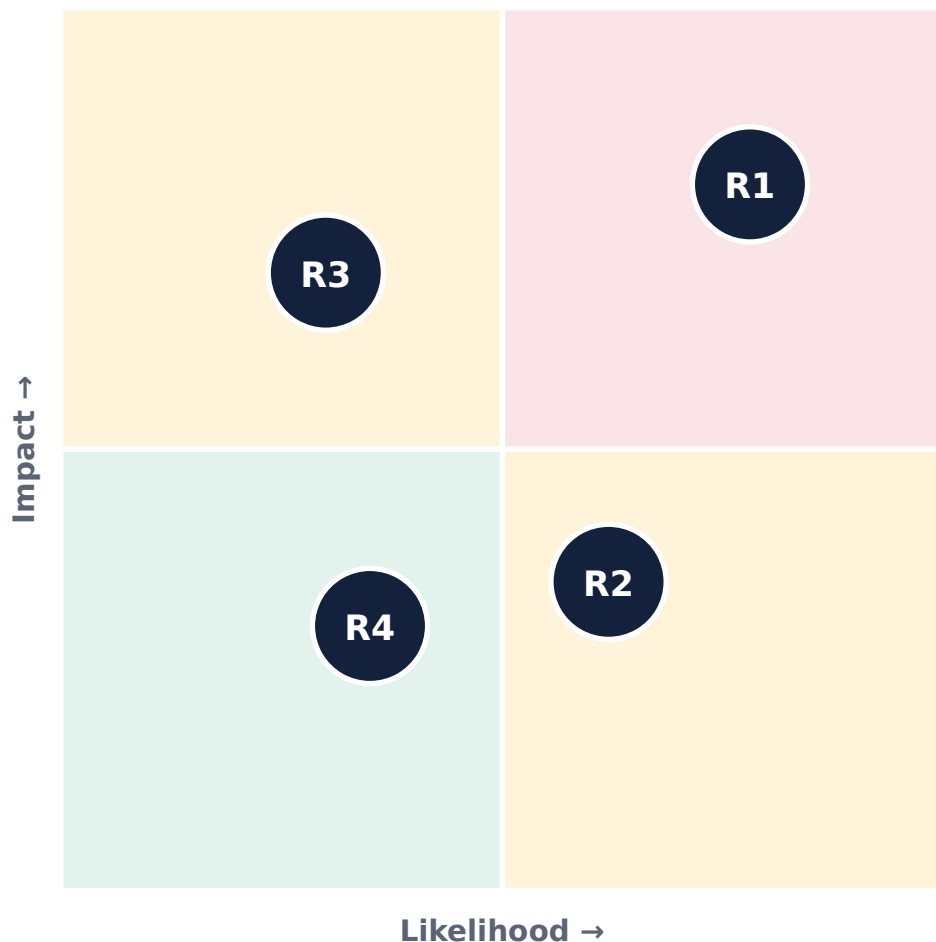


■ ALLOW

■ CONFIRM

■ BLOCK / re-route

Risks & Assumptions



R1 · Speech-to-text drops the negation completely

Mitigation: for irreversible actions, low confidence on a short word, or a missing word where one is expected, sends the call to CONFIRM

R2 · The plug-in adds latency or friction

Mitigation: fast offline path (< 5 ms), LLM only when unclear, and questions only for risky actions

R3 · Dialects and spellings (vendaam, venaam, matt)

Mitigation: lexicon plus LLM fallback, and every human override is added to the training set

R4 · Assumption: callers accept a read-back

In their own language. We validate with pilot CSAT, and an English read-back is the fallback



“Don't cancel” should mean don't cancel.

One plug-in between your AI agent and your APIs. It understands mixed-language callers, confirms when unsure, and audits every action.

OUR ASK

A Freshworks sandbox with Freddy AI Agent + Freshcaller for a 3-brand pilot

LIVE TODAY

codemix-skill.vercel.app: the Round-1 engine this builds on